Kavuri Venkata Sai Satya Sravya Pravallika
NUID:12627628

Project 2 Image Processing Report

1. Introduction

This project aims to predict a **continuous clinical severity score** from 2D hand and wrist X-ray images. Unlike classification, regression requires the network to understand subtle radiographic differences such as:

- joint spacing irregularities
- bone erosion patterns
- density changes
- asymmetry across left/right hand structures

Capturing these fine-grained structural cues demands a highly capable architecture.

The provided baseline (SimpleCNN) served as a starting point, but its shallow depth, lack of pretrained features, and absence of attention made it insufficient for learning complex radiographic patterns.

Therefore, the core contribution of my Project 2 is the design, training, and optimization of a **strong Student Model** using:

- ResNet34 backbone (pretrained)
- CBAM attention modules
- Multi-scale feature fusion
- A deeper fully connected regressor
- AdamW optimizer and cosine LR scheduling

This Student Model delivered substantial performance gains, proving the effectiveness of architectural improvements for medical regression tasks.

Base Model:

The baseline implemented in the starter code was a **SimpleCNN**, a shallow convolutional architecture with only a few layers and no pretrained weights. Such a model is useful for demonstrating the workflow but is fundamentally limited when dealing with complex medical imaging tasks.

Running the Baseline

I first trained the baseline model using the provided script:

```
python main.py --model simple_cnn --epochs 50 --batch_size 32 --lr 0.001
```

This helped me verify that the training pipeline worked correctly before developing the Student Model.

Baseline Results

After training, the baseline model achieved the following metrics:

Metric	Value
MAE	26.85
RMSE	34.01
R^2	0.4336

These results show that the baseline SimpleCNN struggled to capture the subtle radiographic patterns in hand/wrist X-rays. Because it lacks depth, pretrained weights, and attention mechanisms, the model is unable to extract features relating to joint spacing, bone shape, and subtle abnormalities that correlate with severity scores.

The relatively low R^2 and high errors clearly demonstrated that a **more powerful architecture was required**, which motivated the design of my improved Student Model.

2. Student Model

First, rather than simply widening or deepening the SimpleCNN, I chose to build on top of a **pretrained backbone (ResNet34)**. Pretrained models are known to extract far richer features from X-ray images because they carry strong prior knowledge from ImageNet, making them significantly more effective than a CNN trained from scratch on a small medical dataset.

To further improve representational power, I incorporated **attention mechanisms—specifically CBAM (Convolutional Block Attention Module)**—onto deeper layers of the ResNet encoder. These attention blocks helped the model decide *what* features are important (channel attention) and *where* to focus in the image (spatial attention). For medical images, where abnormalities can be subtle, the impact of attention was especially strong.

Additionally, I followed modern architectural strategies by using residual connections throughout the ResNet backbone. Residual learning improves gradient flow, stabilizes training, and helps the model learn deeper hierarchical representations that capture bone density differences, joint spacing variations, and structural deformities.

To prevent overfitting, I also used several regularization techniques, including BatchNorm, Dropout in the regression head, and Weight Decay (L2 regularization) during optimization. These methods helped the model generalize better, especially given the small size of the dataset.

By combining pretrained feature extraction, attention mechanisms, residual blocks, and targeted regularization, the final Student Model became significantly more powerful and achieved a large performance improvement over the baseline.

2.1 ResNet34 Backbone (Pretrained on ImageNet)

ResNet34 was selected as the encoder because it provides a strong balance of depth, computational efficiency, and feature extraction quality.
Its residual connections help the network learn stable representations without vanishing gradients, which is important for regression on subtle medical images.
Using pretrained ImageNet weights allows the model to start with filters already capable of detecting textures, edges, and intensity transitions.
This is especially valuable for hand/wrist radiographs, where severity is determined by fine texture changes and structural variations that a CNN trained from scratch would struggle to learn effectively.

2.2 CBAM Attention Module (Applied to Layer3 and Layer4)

One of the key architectural improvements in my Student Model is the inclusion of CBAM (Convolutional Block Attention Module).
Since medical X-rays often contain very subtle abnormalities—such as narrow joint gaps, bone erosion, or mild deformities—CBAM helps the model emphasize the most informative visual regions.

Channel Attention identifies which feature channels carry medically relevant information, such as abnormal density or unusual bone patterns.
Spatial Attention highlights *where* in the image those abnormalities are located, making it easier for the network to concentrate on clinically important regions.

The combined effect of CBAM improved the model's ability to focus on pathology-related cues, stabilized training, and significantly enhanced regression accuracy.
This was especially important given the dataset's high variability and the subtle nature of the severity indicators.

2.3 Multi-Scale Feature Fusion

ResNet's deeper layers capture different types of information, and both are useful for severity prediction.
Layer3 retains mid-level structural features such as bone alignment and joint shape, while Layer4 captures more abstract global patterns like overall bone condition or deformation.

To leverage both types of information, I applied CBAM to each layer, reduced the outputs into 128-dimensional vectors, and fused them by concatenation.
This multi-scale fusion lets the model combine local fine details with global structural cues, which is critical because medical severity scores depend on both.
This strategy improved the model's ability to capture subtle radiographic features across multiple levels of abstraction.

2.4 Fully Connected Regression Head

After extracting and fusing multi-scale features, a deeper regression head was used to map this representation to a continuous severity score.
The regressor includes BatchNorm for stable gradient flow, Dropout to reduce overfitting, and ReLU activations to introduce nonlinearity.
Using multiple fully connected layers helps the model learn complex relationships between the extracted radiographic features and the final severity score.
This design is essential because regression tasks require more expressive and flexible mappings than simple classification.

5. Training the Student Model

During the first training run, the model showed some fluctuations in the early epochs. This is expected when using attention modules like CBAM because attention layers often require several iterations to stabilize their weighting patterns. After the initial instability, the model converged smoothly and produced strong results.

Initial Student Model Results

Metric	Value
MAE	19.01
RMSE	25.78
R^2	0.674

These results already showed a significant improvement over the SimpleCNN baseline. Even without any hyperparameter tuning, the Student Model benefited from deeper features, attention mechanisms, and pretrained initialization, which allowed it to capture more meaningful radiographic patterns.

6. Hyperparameter Optimization

After establishing a strong baseline with the Student Model, I performed systematic tuning to further refine performance. Each hyperparameter was tested individually and in combination to understand its impact on model behavior.

Learning Rates

- **1e-3** — too aggressive; caused unstable or noisy training
- **5e-4** — more stable and consistently reached lower validation losses

- **Cosine decay schedule** — provided smooth LR reduction that improved late-epoch convergence

A moderately low learning rate paired with cosine annealing produced the most reliable training dynamics.

Batch Sizes

- **32 → too unstable** due to gradient noise and small dataset size
- **16 → optimal**, giving the best balance between stability and model generalization

Larger batches flattened out important gradient signals, while batch size 16 helped maintain meaningful variance.

Optimizers

- **Adam → better**; adapted learning rates improved early optimization
- **AdamW → best stability + regularization**; decoupled weight decay helped prevent overfitting

AdamW consistently produced smoother learning curves and higher R^2 values.

Schedulers

- **ReduceLROnPlateau → inconsistent** with this regression task; learning rate would drop too late or too early
- **Cosine Annealing → smooth convergence** and helped avoid premature LR drops

Cosine scheduling worked especially well with AdamW, supporting a more stable descent toward the final minima.

Regularization

- **weight_decay = 1e-5** significantly improved generalization by preventing the model from overfitting to noise in the X-ray images
- Helped maintain lower validation MAE in later epochs

Overall, the combination of **AdamW + Cosine LR + weight decay + batch size 16** produced the most stable and best-performing configuration.

7. Final Best Model (Optimized Student Model)

Final command:

```
python main.py \
  --model student \
  --epochs 50 \
  --batch_size 16 \
  --lr 5e-4 \
  --optimizer adamw \
  --scheduler cosine \
  --weight_decay 1e-5
```

Final Results

Metric	Final	Baseline	Improvement
MAE	16.82	26.85	↓ 37%
RMSE	23.07	34.00	↓ 32%
R^2	0.7392	0.4336	↑ +0.30

The Student Model clearly outperformed the baseline, validating the architectural decisions.

8. Inference Results

Predicted: 8.81 | True: 3.00

The model shows strong trend-following behavior but struggles slightly with very extreme or very low scores—common in medical regression tasks with imbalanced distributions.

Validation set statistics:

- **Min prediction:** 6.81
- **Max prediction:** 171.63
- **Mean:** 37.91
- **Std:** 38.05

The predicted distribution matches the real severity distribution well.

---

9. Key Insights & Learnings

1. Using pretrained ResNet34 massively improves medical regression tasks.
2. CBAM attention helps the model focus on radiographically meaningful areas.
3. Multi-scale feature fusion was crucial — combining layer3 and layer4 features improved performance significantly.
4. AdamW + Cosine LR scheduling stabilized training and improved convergence speed.
5. Batch size 16 provided the best gradient noise balance.
6. Regression requires deeper feature understanding than classification, especially on medical images with subtle abnormalities.

10. Conclusion

This project demonstrates how architectural choices and systematic experimentation can dramatically improve regression performance on medical X-ray datasets.

My Student Model:

- Uses a strong pretrained backbone
- Enhances feature extraction through attention
- Merges multi-level representations
- Stabilizes regression with a deeper MLP head
- Achieves significantly better metrics than the baseline

Final performance:

MAE = 16.82

RMSE = 23.07

R^2 = 0.7392

This confirms that deep, attention-based models with multi-scale fusion are much better suited for medical regression tasks than traditional CNNs.

Google Colab code (Scripts)

https://colab.research.google.com/drive/1oBt1ItjRVAXTvO9Bg08UzW_aHDoGaaus?usp=sharing